

22683

23242

3 Hours / 70 Marks

Seat No.

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- Instructions :**
- (1) All Questions are *compulsory*.
 - (2) Answer each next main Question on a new page.
 - (3) Illustrate your answers with neat sketches wherever necessary.
 - (4) Figures to the right indicate full marks.

Marks

1. Attempt any FIVE of the following :

10

- (a) Define lazy learners & eager learners.
- (b) What is feature scaling ?
- (c) State any four applications of random forest.
- (d) How to select value of 'K' in K-nearest neighbour Algorithm ?
- (e) State K-Means clustering.
- (f) List the methods which are used to find hyper parameter.
- (g) Define GPT (Generative Pre-trained Transformer).

2. Attempt any THREE of the following :

12

- (a) Describe process of feature engineering.
- (b) Demonstrate working of decision tree algorithm with attribute selection measures.
- (c) Describe dimensionality reduction with example.
- (d) Describe activation function in neural network.



3. Attempt any THREE of the following : 12

- (a) List any two advantages and disadvantages of support vector machine.
- (b) Describe Principle Component Analysis (PCA) method for dimensionality reduction.
- (c) With the suitable example, explain how Bayes Theorem is applied.
- (d) Give classifications of RNN & CNN.

4. Attempt any THREE of the following : 12

- (a) Explain hold out method for model selection.
- (b) Describe two types of ensemble learning methods.
- (c) Explain the types of support vector machines.
- (d) Explain common use cases of backpropagation ANN.

5. Attempt any TWO of the following : 12

- (a) Consider following training dataset of weather, apply naive bayes classification on given data and solve the given problem statement :

Weather	S	O	R	S	S	O	R	R	S	R	S	O	O	R
Play	N	Y	Y	Y	Y	Y	N	N	Y	Y	N	Y	Y	N

S – Sunny

N – No

O – Overcast

Y – Yes

R – Rainy

Fig : Sample dataset

Problem statement :

- (i) Player will play if weather is sunny. Is this statement correct ?

- (b) Implement python code for given steps to implement KNN algorithm :
 - (i) Import suitable modules (import kneighbor classification)
 - (ii) Create dataset
 - (iii) Visualize dataset
 - (iv) Splitting data into training and testing set
 - (v) KNN classifier implementation
 - (vi) Prediction for KNN classifier
- (c) Describe CNN. Write steps to build an image classification model using CNN.

6. Attempt any TWO of the following :

12

- (a) Describe ANN with its applications.
 - (b) Describe three approaches of transfer learning.
 - (c) Write python code to implement K-means clustering for given steps :
 - (i) Import libraries
 - (ii) Prepare sample data
 - (iii) Choose no. of cluster (K)
 - (iv) Fit data to algorithm
 - (v) Get cluster center
 - (vi) Get labels
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